FPGA-Based RTL Implementation and Emulation of the BB84 Quantum Key Distribution Protocol

1. Abstract

This project presents a comprehensive Register-Transfer Level (RTL) implementation and digital emulation of the BB84 Quantum Key Distribution (QKD) protocol using Verilog HDL on FPGA platforms. The objective is to design a hardware-oriented architecture capable of simulating the secure exchange of cryptographic keys through digital representations of polarized photon states. The system will employ Finite State Machines (FSMs), pseudo-random basis generation, protocol-level synchronization, and Quantum Bit Error Rate (QBER) monitoring to emulate the behavior of BB84 communication.

The project focuses on the digital control and protocol-processing layers used in modern FPGA-assisted QKD systems rather than implementing physical quantum optical hardware. Functional verification and timing analysis will be conducted using simulation testbenches and FPGA synthesis tools to evaluate performance, resource utilization, and intrusion detection capability.

2. Background, Introduction and Problem Statement

Conventional encryption systems suchas RSA relyon the computational difficultyof solvingmathematical

problems, particularly integer factorization and discrete logarithms. The emergence of large-scale quantum computers and algorithms such as Shor’s Algorithm threatens the long-term security of these systems by significantly reducing the complexity required to recover private keys.

Quantum Key Distribution (QKD) introduces a fundamentally different approach to secure communication by utilizing the physical properties associated with quantum-state measurements. Among the most widely studied QKD schemes is the BB84 protocol, which enables two parties to establish a shared secret key while detecting eavesdropping attempts through error analysis.

Practical QKD systems require high-speed digital control, synchronization, and post-processing units capable of operating in real time. FPGA technology provides an ideal platform for implementing such control architectures due to its parallel processing capability, deterministic timing behavior, and hardware-level configurability.

This project addresses the engineering challenge of designing an FPGA-based RTL architecture capable of digitally emulating the operational behavior of the BB84 protocol. The work focuses on protocol control, basis selection, sifting logic, and QBER analysis while excluding the implementation of physical optical components such as photon emitters or detectors.

3. Literature Review

Thissection evaluatescontemporarymethodologies for implementing Quantum Key Distribution (QKD) protocols on FPGA platforms. Implementing QKD requires a rigorous approach to hardware-accelerated post-processing and digital logic design to meet real-time performance requirements.

Comparative Study of FPGA-based QKD Implementations

Paper Reference	FPGA Platform	HDL	Protocol/ Algorithm	Key Performance Metrics
[2]	Xilinx Virtex-7 // Intel/Terasic DE10-Standard	Not Specified	CV-QKD / LDPC	100.90 M symbols/s throughput
[3]	Xilinx Virtex-7 // Intel/Terasic DE10-Standard	VHDL	BB84 / PRINCE	3.931 Gbps throughput; 7.29 Mbps/ slice
[4]	Xilinx Virtex-5 // Intel/Terasic DE10-Standard	VHDL	BB84	High-speed bit coding; flexible integration

[2] Analysis (Yang et al., 2020)

This study focuses on high-speed post-processing for Continuous-Variable QKD (CV-QKD). The authors utilized a Xilinx Virtex-7 (XC7VX690T) to implement parallel LDPC decoders and privacy amplification. By optimizing the sum-product algorithm and employing a modified LDPC construction algorithm, they achieved a remarkable throughput of 100.90 M symbols/s, demonstrating the superior parallel processing capability of FPGAs compared to CPU/GPU implementations.

[3] Analysis (Abdullah & Obeid, 2021)

This research proposes an efficient hardware architecture for the PRINCE block cipher integrated with the BB84 protocol. Implementing the design on Virtex-4 and Kintex-7 platforms using VHDL, the authors achieved significant optimization in latency and throughput. Specifically, the Kintex-7 implementation yielded a throughput of 3.931 Gbps and an efficiency of 7.29 Mbps/slice, validating the integration of cryptographic algorithms within the QKD framework on FPGA hardware.

[4] Analysis (Li et al., 2016)

Focusing on the fundamental implementation of the BB84 protocol, this study utilized a Virtex-5 platform with VHDL to design a digital control system. The design covers essential sub-modules including random data generation, laser source driving, and error rate estimation. This work confirms the feasibility of FPGA-based digital control for QKD, highlighting advantages such as compact size, high bitrates, and ease of algorithm updates.

Synthesis

Theliterature demonstrates a clear evolutionary trend in FPGA-based QKD. While early implementations [4] focused on the basic digital control of the BB84 optical setup, modern research [2, 3] has shifted towards high-throughput post-processing and the integration of advanced cryptographic primitives (such as the PRINCE algorithm) to ensure end-to-end security and real-time operational efficiency.

The reviewed studies collectively demonstrate that FPGA architectures are highly suitable for implementing the digital layers of QKD systems, particularly in applications requiring deterministic timing, high throughput, parallel processing, and hardware-level synchronization.

4. Project Rationale

The primary motivation is to apply advanced principles of Digital Systems Design and Computer Architecture through the implementation of a complex hardware-oriented cryptographic protocol. The project emphasizes RTL design methodologies including datapath construction, control-unit design, Finite State Machines (FSMs), modular hardware integration, and timing verification.

Additionally, the project provides practical experience in hardware security, FPGA-based acceleration, and protocol-level digital emulation of quantum communication systems. The work demonstrates how concepts associated with QKD protocols can be translated into synthesizable digital architectures suitable for FPGA implementation.

5. Aims and Objectives

The project aims to design and verify a complete FPGA-based RTL architecture for digital emulation of the BB84 protocol.

The primary objectives include:

- Designing an FSM-controlled Alice module responsible for generating random data bits and basis selections.
- Designing a Bob receiver module capable of random basis selection and photon-state measurement emulation.
- Implementing an Eve module to emulate eavesdropping attacks and analyze their effect on QBER.
- Designing a Quantum Channel Emulator to transfer encoded photon states between modules.
- Implementing Sifting Logic for basis comparison and valid key extraction.
- Implementing a QBER Analysis Unit to detect abnormal error rates.
- Evaluating FPGA resource utilization and timing performance.
- Verifying the architecture using simulation testbenches and FPGA synthesis tools.

6. Scope of the Project

ThescopeincludesRTL design,Verilog implementation, FPGA synthesis, functional verification, and performance analysis of a digital BB84 protocol emulator.

The project focuses exclusively on the digital control and protocol-processing layers of QKD systems and does not include physical optical quantum hardware such as lasers, photon emitters, or optical detectors. The architecture consists of the following major modules:

Alice Module

An FSM-based transmitter unit responsible for:

- Generating pseudo-random data bits.
- Generating pseudo-random basis selections.
- Encoding logical photon states.
- Transmitting encoded symbols through the Quantum Channel Emulator.

Bob Module

An FSM-based receiver unit responsible for:

- Generating pseudo-random measurement bases.
- Measuring received photon-state representations.
- Recovering data bits.
- Forwarding basis information to the sifting stage.

Eve Module

An optional attack module used to emulate eavesdropping behavior by intercepting transmitted photon-state representations and performing intermediate measurements that introduce detectable errors into the communication process.

Quantum Channel Emulator

A digital communication layer responsible for transferring encoded photon-state representations between Alice and Bob.

The logical photon states are represented using 2-bit encoded symbols:

- 00 → Horizontal Polarization
- 01 → Vertical Polarization
- 10 → Diagonal Polarization (/)
- 11 → Anti-Diagonal Polarization (\)

Sifting Logic

Adedicatedprocessing module responsible for basis comparison between Alice and Bob. Only bits generated using matching bases are retained as part of the shared secret key, while mismatched measurements arediscarded.

QBER Calculation Unit

Amonitoringmoduleresponsiblefor calculating the Quantum Bit Error Rate (QBER):

QBER = Number of Incorrect Shared Bits / Total Shared Bits

If the calculated QBER exceeds a predefined threshold, the communication session is considered compromiseddueto possible eavesdropping activity.

PRNG Unit

Theprojectemploys FPGA-compatible Pseudo-Random Number Generators (PRNGs), such as Linear Feedback Shift Registers (LFSRs), for protocol-level digital simulation. These generators are used solely for hardware emulation purposes and do not claim physical quantum randomness.

Deliverables include:

- Verilog HDL source code.
- FSM diagrams.
- Datapath and block diagrams.
- Testbenches and simulation waveforms.
- FPGA synthesis and utilization reports.

7. Proposed Methodology

The project follow sahierarchical Top-Down RTL Design methodology.

Design Phase

- Development ofblock diagrams and datapath architectures.
- FSM state modeling for Alice, Bob, and Controller modules.
- Definition of communication interfaces between protocol units.
- Definition of encoded photon-state representations.

Development Phase

Implementationofhardwaremodules using Verilog HDL, including:

- Alice FSM
- Bob FSM
- Eve Module
- Quantum Channel Emulator
- Sifting Logic
- QBER Analyzer
- PRNG Modules
- ControllerFSM

Integration Phase

Integrationand synchronization of all hardware modules using clocked sequential logic and shared communication buses.

Data transfer between modules follows the architecture below:

Alice → QuantumChannelEmulator → Eve (optional) → Bob → Sifting Logic → QBER Analyzer

Verification Phase

Simulationtestbenches will bedeveloped to verify:

- Correct protocol execution.
- Basis matching behavior.
- Key generation functionality.
- Eavesdropping detection.
- QBER variation under attack conditions.

Timing simulationsand waveform analysis will be conducted using FPGA development tools.

8. Proposed Solution and Anticipated Results

The proposed solution is a modular, FPGA-based hardware architecture capable of digitally emulating the operational behavior of the BB84 QKD protocol.

The system is expected to:

- Successfully generate identical shared keys between Alice and Bob under normal operating conditions.
- Detect eavesdropping attempts through increased QBER values.
- Demonstrate stable synchronization between communicating modules.
- Achieve efficient FPGA resource utilization.
- Demonstrate scalable RTL architecture suitable for future hardware expansion.

Under normal communication conditions, the QBER is expected to remain below the predefined security threshold. When the Eve module is enabled, the system should demonstrate a measurable increase in QBER, leading to rejection of compromised keys.

The final implementation is expected to validate the suitability of FPGA technology for high-speed digital emulation and control of QKD protocols.

9. References

[1] Harris, D., & Harris, S. (2012). Digital Design and Computer Architecture. Morgan Kaufmann.

[2] Yang, S. S., Lu, Z. G., & Li, Y. M. (2020). High-Speed Post-Processing in Continuous-Variable Quantum Key Distribution Based on FPGA Implementation. Journal of Lightwave Technology
http://ioe.sxu.edu.cn/lab10/papers/JOURNAL%20OF%20LIGHTWAVE%20TECHNOLOGY%2038,%203935.pdf

[3] Abdullah, A. A., & Obeid, N. R. (2021). Efficient Implementation for PRINCE Algorithm in FPGA Based on the BB84 Protocol. Journal of Physics: Conference Series, 1818(1), 012216. DOI: 10.1088/1742-6596/1818/1/012216
https://iopscience.iop.org/article/10.1088/1742-6596/1818/1/012216

[4] Li, H., Zhang, M., & Wang, K. (2016). The FPGA Implementation of Quantum Key Distribution Based on BB84 Protocol. 2nd Workshop on Advanced Research and Technology in Industry Applications (WARTIA 2016).
https://www.researchgate.net/publication/303861706_The_FPGA_Implementation_of_Quantum_Key_Distribution_Based_on_BB84_protocol

[5] OpenQKDsecurity: Open-source framework for security analysis of QKD protocols. GitHub Repository.

[6] Benchmarking FPGA-based QKD Accelerators. arXiv (2024).

[7] A Survey on Quantum Key Distribution: Protocols, Hardware Implementations and Future Directions. IET Quantum Communication.

[8] https://www.scirp.org/journal/paperinformation?paperid=54583 ,
https://ietresearch.onlinelibrary.wiley.com/doi/full/10.1049/qtc2.70025 ,
https://arxiv.org/html/2411.19880v1

[9] Implementation-of-Quantum-Key-Distribution-protocol-BB84-on-FPGA GitHub Repository
https://github.com/devchadha-jmi/Implementation-of-Quantum-Key-Distribution-protocol-BB84-on-FPGA